Reference oracle	HR–BC	BR–BC	HR–HC
Train HR–BC (all train)	0.92 ± 0.02	0.10 ± 0.07	–
Train HR–HC (all train)	0.47 ± 0.09	0.28 ± 0.13	0.93 ± 0.01
Gold HR–BC (XSTest)	0.70 ± 0.11	0.39 ± 0.11	0.68 ± 0.05
Gold HR–HC (XSTest)	0.51 ± 0.03	0.24 ± 0.13	0.81 ± 0.02

Table 10: Mean cosine similarity (mean $\pm$ std over random resamplings where applicable) between various dataset-specific directions and different oracle refusal directions at layer 20, position -2. HR–BC and HR–HC directions are highly stable and align well with their corresponding oracles, while BR–BC remains clearly separated.

picture in which HR–BC and HR–HC form stable, well-separated refusal-related directions that are distinct from the BR–BC axis and from other harm/compliance contrasts.

D.1 Experiment 5: Hook Variants and Robustness

Goal. Check whether the causal effect of SAE refusal latents depends delicately on where and how we hook, or whether it is robust across reasonable choices of layer and token pattern.

Summary. We vary (i) the SAE layer (9/20/31), (ii) whether the steering direction is added at the prompt-only, during generation-only, or at both, and (iii) how often we inject it during generation (every token vs. every second or third token). Across these settings, we observe a smooth trade-off: more frequent or deeper hooks require smaller α to achieve similar HR/BR behavior, but the overall shape of the curves is essentially unchanged. This mirrors the activation-space results and suggests that the SAE-based refusal directions are not a fragile artefact of a single hook choice.

D.2 Experiment 6: Taxonomy of Refusal Latents

Goal. Move from "refusal latents exist and are causal" to "*what* do these latents represent?" by building a lightweight semantic taxonomy of the most refusal-associated latents.

Method (high level). For a fixed SAE layer, we take the top refusal-moving latents across several splits and, for each latent, collect its highest-activating prompt–response pairs. We then use an external LLM to summarise common patterns per latent and assign short labels, and a second pass to group latents into a small number of higher-level categories.

Findings. Latents naturally cluster into a handful of recurring themes, such as:

- *self-harm and crisis help* (explicit refusals that redirect to support resources),
- *sexual and relationship boundaries*,
- *crime, fraud, and physical harm*,
- *copyright / full-text reproduction*, and
- *professional and legal disclaimers*.

Within these themes we see finer-grained subtypes, e.g. features that emphasise "I am just an AI and have no body" vs. features that foreground illegality or lack of professional qualifications. This aligns closely with the stylistic patterns illustrated in Appendix A: different refusal directions lean on different subsets of these latent themes when framing their refusals.

A complementary run on a *combined* HR/BC pool confirms that these themes are not just artifacts of analyzing each dataset in isolation. We construct a balanced mixture by downsampling XSTest, WildGuard-Mix, CoCoNot, and SorryBench to the size of the smallest dataset and then balancing refusals vs. non-refusals, and rerun the same latent-mining and GPT-labelling pipeline on this pooled set. The top refusal-moving latents at a late layer again collapse into a small set of macro-classes: (i) hate/abuse, defamation, and disinformation; (ii) violence, illicit instructions, and sexual exploitation; (iii) privacy/PII, copyright, and impossible or omniscient knowledge; and (iv) capability and access limits, including multimodal/tool-access gaps and anthropomorphic self-history requests. That these same macro-classes emerge when all datasets are mixed suggests that the taxonomy is capturing a compact, reusable backbone of refusal-related features that is shared across benchmarks, with dataset-specific refusal styles largely reflecting different weightings over this common feature set rather than wholly distinct mechanisms.

E Latent Semantic Annotation

To semantically interpret the SAE latents associated with refusal behavior, we developed a two-round LLM-assisted annotation pipeline. First, we identified the top-10 latents exhibiting the highest refusal-moving scores from our contrastive analysis between refusal and compliance examples. For each selected latent, we extracted the 20 prompts

Split	$\alpha = 0$			$\alpha = 0.25$			$\alpha = 0.5$			$\alpha = 0.75$			$\alpha = 1.0$		
	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.710	0.44	0.26	0.705	0.44	0.25	0.720	0.45	0.25	0.755	0.48	0.29	0.750	0.55	0.31
Incomplete-CCN	0.710	0.44	0.26	0.710	0.44	0.26	0.705	0.44	0.25	0.720	0.44	0.28	0.750	0.46	0.30
Indeterminate-CCN	0.710	0.44	0.26	0.710	0.44	0.28	0.715	0.47	0.28	0.745	0.46	0.33	0.745	0.49	0.34
Safety-CCN	0.710	0.44	0.26	0.710	0.45	0.25	0.745	0.47	0.30	0.765	0.55	0.34	0.755	0.57	0.32
Unsupported-CCN	0.710	0.44	0.26	0.710	0.44	0.26	0.700	0.45	0.27	0.735	0.48	0.31	0.745	0.52	0.31
CrimeAssistance-SB	0.710	0.44	0.26	0.715	0.45	0.26	0.740	0.46	0.30	0.760	0.53	0.33	0.745	0.60	0.31
HateSpeech-SB	0.710	0.44	0.26	0.715	0.45	0.26	0.735	0.46	0.31	0.755	0.53	0.32	0.755	0.56	0.33
Inappropriate-SB	0.710	0.44	0.26	0.715	0.46	0.25	0.725	0.49	0.30	0.760	0.52	0.32	0.720	0.60	0.30
Advice-SB	0.710	0.44	0.26	0.710	0.45	0.25	0.750	0.46	0.30	0.770	0.54	0.32	0.735	0.58	0.31
SafetyCore-WGM	0.710	0.44	0.26	0.710	0.46	0.26	0.730	0.49	0.29	0.765	0.57	0.32	0.745	0.58	0.31
OverRefusal-XST	0.710	0.44	0.26	0.715	0.45	0.26	0.745	0.46	0.29	0.765	0.52	0.33	0.740	0.57	0.31

Table 11: Performance of Llama models steered along 11 refusal directions on the controlled test set, reported in terms of overall accuracy $\big($Acc = (HR' + BC')/200$\big)$, refusal rate $\big($RR = HR'/100$\big)$, and over-refusal rate $\big($ORR = BR'/100$\big)$, where R' and C' refer to the WildGuard judgement of whether the steered response is a refusal or a compliance, respectively. Each split contains 50 harmful-refusal, 50 harmful-compliance, 50 benign-compliance, and 50 benign-refusal prompts.

with the highest activation values and tokenized them such that the most strongly activating tokens were marked with brackets to highlight the key triggering patterns. These annotated examples were then passed to GPT-4, which generated an initial semantic label, description, and characteristic patterns for each latent based on the bracketed activation contexts (Round 1: Initial Annotation). Next, the model received all latent annotations together and was prompted to organize them into a maximum of four coherent semantic categories, identifying commonalities and distinctions across latents (Clustering Stage). Finally, GPT-4 was provided with each latent's original description alongside its assigned cluster description and asked to produce refined labels that better reflect the latent's role within its category (Round 2: Category-Aware Refinement). Crucially, this annotation pipeline was executed under multiple settings: once on a combined dataset (mixing XSTest, WildGuard, CoCoNot, and SorryBench, balanced to 1,320 examples) and separately on individual dataset subcategories (e.g., CoCoNot's "Humanizing requests," "Unsupported requests," or SorryBench's "Hate speech generation," "Assistance with crimes or torts"). By annotating the same latents across different data contexts, we reveal their multi-faceted nature—a single latent may receive distinct semantic interpretations depending on which examples most strongly activate it in a given setting. For instance, latent 550 is labeled "Commissioned public-harm communications" in the combined analysis focusing on hate speech and defamation, but receives the label "Unsupported modality request" when analyzed on CoCoNot's capability-focused examples. Similarly, latent 7137 is interpreted as "Missing-input/external-content requests" in the combined setting, but as "Malware and doxxing requests" in SorryBench's crime-focused subcategory and "Translate/Transcribe Attached Media" in CoCoNot's unsupported requests. This polysemous behavior suggests that refusal-related latents encode context-sensitive features rather than fixed categorical detectors.

F Refusal Latent Overlap between Splits

We show in Table 16 the overlap of top-3000 refusal latents across all the layers between the directions of different splits, while Tables 17,18 and 19 show the overlap between the top-1000 latents per split for the layer 9, 20, 31, respectively.

Split	$\alpha = 10$			$\alpha = 30$			$\alpha = 60$			$\alpha = 100$		
	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.855	0.95	0.24	0.785	0.95	0.38	0.670	1.00	0.66	0.515	1.00	0.97
Incomplete-CCN	0.870	0.94	0.20	0.770	0.92	0.38	0.735	0.94	0.47	0.550	0.98	0.88
Indeterminate-CCN	0.860	0.94	0.22	0.780	0.96	0.40	0.635	0.98	0.71	0.545	0.99	0.90
Safety-CCN	0.850	0.97	0.27	0.715	0.99	0.56	0.515	0.99	0.96	0.500	1.00	1.00
Unsupported-CCN	0.835	0.89	0.22	0.765	0.91	0.38	0.630	0.95	0.69	0.515	0.99	0.96
CrimeAssistance-SB	0.865	0.95	0.22	0.660	0.98	0.66	0.500	1.00	1.00	0.500	1.00	1.00
HateSpeech-SB	0.860	0.94	0.22	0.665	0.98	0.65	0.500	1.00	1.00	0.500	1.00	1.00
Inappropriate-SB	0.860	0.93	0.21	0.695	0.99	0.60	0.505	1.00	0.99	0.500	1.00	1.00
Advice-SB	0.825	0.94	0.29	0.580	0.98	0.82	0.500	1.00	1.00	0.500	1.00	1.00
SafetyCore-WGM	0.850	0.95	0.25	0.690	0.97	0.59	0.505	1.00	0.99	0.500	1.00	1.00
OverRefusal-XST	0.860	0.96	0.24	0.650	0.95	0.65	0.520	1.00	0.96	0.510	1.00	0.98

Table 12: Performance of gemma model steered along 11 refusal directions on the controlled test set, reported in terms of overall accuracy $\big($Acc = (HR' + BC')/200$\big)$, refusal rate $\big(RR = \frac{\mathrm{TP}}{\mathrm{TP+FN}}\big)$, and over-refusal rate $\big(ORR = \frac{\mathrm{FP}}{\mathrm{FP+TN}}\big)$, where TP, TN, FP, FN are computed using WildGuard judgements of whether the steered response is a refusal or a compliance. The unsteered base model ($\alpha = 0$) attains 50% on all three metrics.

Split	$\alpha = 1.0$			$\alpha = 1.1$			$\alpha = 1.2$		
	Acc	RR	ORR	Acc	RR	ORR	Acc	RR	ORR
Humanizing-CCN	0.485	0.560	0.590	0.490	0.660	0.680	0.435	0.680	0.810
Incomplete-CCN	0.500	0.970	0.970	0.465	0.920	0.990	0.410	0.770	0.950
Indeterminate-CCN	0.435	0.850	0.980	0.435	0.850	0.980	0.440	0.880	1.000
Safety-CCN	0.485	0.970	1.000	0.480	0.960	1.000	0.470	0.940	1.000
Unsupported-CCN	0.500	0.960	0.960	0.495	0.950	0.960	0.495	0.970	0.980
CrimeAssitance-SB	0.370	0.600	0.860	0.285	0.550	0.980	0.305	0.590	0.980
HateSpeech-SB	0.490	0.980	1.000	0.495	0.990	1.000	0.450	0.860	0.960
Inappropriate-SB	0.490	0.970	0.990	0.470	0.930	0.990	0.480	0.950	0.990
Advice-SB	0.485	0.960	0.990	0.475	0.920	0.970	0.415	0.830	1.000
SafetyCore-WGM	0.485	0.970	1.000	0.490	0.960	0.980	0.470	0.940	1.000
OverRefusal-XST	0.550	0.970	0.870	0.530	1.000	0.940	0.510	1.000	0.980

Table 13: Performance of the SAE-steered Llama model (single refusal direction) on the controlled test set, reported in terms of overall accuracy $\big(\mathrm{Acc} = (\mathrm{HR} + \mathrm{BC})/200\big)$, refusal rate $\big(\mathrm{RR} = \mathrm{HR}/100\big)$, and over-refusal rate $\big(\mathrm{ORR} = \mathrm{BR}/100\big)$.